Metals Market Index MMI

MMi Dashboard

Iron Ore Price Indices

Iron Ore Port Stock (FOT Qingdao)
IOPI62 62% Fe Fines RMB/t

1006

4 0.37%

Dec 20th, 2023

Iron Ore Port Stock (FOT Qingdao)
IOPI65 65% Fe Fines RMB/t

1118

3.68 0.33%

Dec 20th, 2023

Iron Port Stock (FOT Qingdao)
IOPI58 58% Fe Fines RMB/t

917

1 0.15%

Dec 20th, 2023

Iron Ore Seaborne (CFR Qingdao)
IOSI62 62% Fe Fines USD/dmt

134.83

2.63 1.99%

Dec 20th, 2023

Iron Ore Seaborne (CFR Qingdao)
IOSI65 65% Fe Fines USD/dmt

158.25

0.00 0.00%

Dec 20th, 2023

Iron Ore Port Stock (FOT Qingdao)
IOPLI 62.5% Fe Lump RMB/t

1165

-18 -1.52%

Week Ending Dec 15th, 2023

Exchange Traded Contracts

DCE Iron Ore 62% Fines
I2401 (Jan) RMB/t (3pm close)

939.00

16.00 1.73%

Dec 20th, 2023 (3pm close)

SGX Iron Ore (CFR Qingdao) 62% Fe Fines
January 24 USD/dmt

133.85

1.45 1.10%

Dec 20th, 2023 (5.30 pm Print)

SHFE Rebar
RB2405 (Jan) RMB/t

4058

133 3.39%

Dec 20th, 2023 (3pm close)

Freight Rates

C3, Tubarao - Qingdao USD/t

24.06

-0.33 -1.34%

Dec 19th, 2023

C5, W. Australia - Qingdao USD/t

13.67

-0.63 -4.37%

Dec 19th, 2023

Steel Price

Steel Rebar (China Domestic) RMB/t

4030

-60 -1.47%

Week Ending Dec 15th, 2023

Inventory Levels

Iron Ore Inventory at Chinese Ports (35)
million tonnes

113.39

-1.71 -1.49%

Week Ending Dec 15th, 2023

Steel Inventory in China
million tonnes

9.63

0.03 0.31%

Week Ending Dec 15th, 2023

Steel Price

Steel HRC (China Domestic) RMB/t

4040

-50 -1.22%

Week Ending Dec 15th, 2023

IRON ORE PORT STOCK INDEX (IOPI)

Dec 20th, 2023		FOT Qingdao (inc. 13% VAT), RMB/wet tonne							CFR Qingdao Equivalent (exc. 13% VAT), USD/dry tonne [1]						
Index	Fe Content	Price	Change	Change %	MTD	YTD	Low [2]	High [2]	Price	Change	Change %	MTD	YTD	Low [2]	High [2]
IOPI62	62% Fe Fines	1006	3.7	0.37%	861	883	770	1023	132.56	0.53	0.4%	114.01	119.85	100.95	134.65
IOPI58	58% Fe Fines	917	1.3	0.15%	765	767	680	924	121.78	0.21	0.2%	101.99	104.49	89.86	122.47
IOPI65	65% Fe Fines	1118	3.7	0.33%	978	1016	881	1135	147.70	0.53	0.4%	129.94	138.63	116.19	149.76

IRON ORE SEABORNE INDEX (IOSI)

Dec 20th, 2023		CFR Qingdao, USD/dry tonne						
Index	Fe Content	Price	Change	Change %	MTD	YTD	Low [2]	High [2]
IOSI62	62% Fe Fines	134.83	2.63	1.99%	114.87	114.87	99.20	146.75
IOSI65	65% Fe Fines	158.25	0.00	0.00%	128.26	128.26	114.90	162.60

MARKET COMMENTARY

DCE iron ore have stopped falling and rebounded, the main contract I2405 closed 939, an increase of 1.46% throughout the day. some traders were active to sell, Some steel mills tended to be wait-and-see, and purchased on demand. today's overall market transaction atmosphere in general. PBF at Shandong port dealt 991-995 yuan/mt; increased 6-10 yuan/mt over yesterday. PBF at Tangshan port dealt 1005-1008 yuan/mt; increased 5-8 yuan/mt over yesterday. The operating rate of blast furnaces according to SMM statistics is 89.74%, a decrease of 0.3% compared to last week. The utilization rate of blast furnace production capacity is 90.42%, an increase of 0.08% compared to the previous period. The daily average iron production of the sample steel plant was 2.1689 million tons, an increase of 0.0600 tons compared to the previous period. However, the overseas shipment of iron ore is still in the year-end surge stage, with supply tending to be loose, demand continuing to weaken, and support for iron ore being weak. But approaching the pre holiday restocking, market expectations are high, and ore prices are prone to rise but difficult to fall.

IRON ORE PORT LUMP INDEX (IOPLI)

Week Ending Dec 15th, 2023		FOT Qingdao (inc. 13% VAT), RMB/wet tonne							CFR Qingdao Equivalent (exc. 13% VAT), USD/dry tonne [3]						
Index	Fe Content	Price	W-o-W	Change %	MTD	YTD	Low [2]	High [2]	Price	W-o-W	Change %	MTD	YTD	Low [2]	High [2]
IOPLI62	62.5% Fe Lump	1165	-18	-1.5%	971	1031	903	1183	147.41	-2.31	-1.54%	122.22	134.93	114.47	147.41

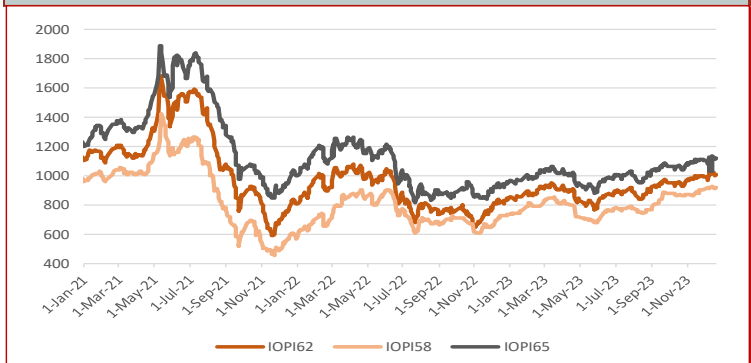
IRON ORE DOMESTIC CONCENTRATE SPOT PRICE ASSESSMENTS AND COMPOSITE INDEX

Week Ending Dec 15th, 2023				RMB/tonne (excluding tax) [3]				USD/tonne (excluding tax) [3]			
Province	Region	Product	Basis	This week	Change %	Low [2]	High [2]	This week	Change %	Low [2]	High [2]
Hebei	Hanxing	66% Fe Concentrate	Dry	1166	1.4%	779	1645	163.99	1.41%	110.31	255.69
Hebei	Qian'an	65% Fe Concentrate	Dry	1205	1.3%	780	1630	169.47	1.28%	110.51	251.57
Liaoning	Anshan	65% Fe Concentrate	Wet	925	2.8%	620	1310	130.09	2.80%	87.40	202.32
Shandong	Zibo	65% Fe Concentrate	Dry	1226	1.5%	800	1752	172.43	1.51%	0.00	272.32

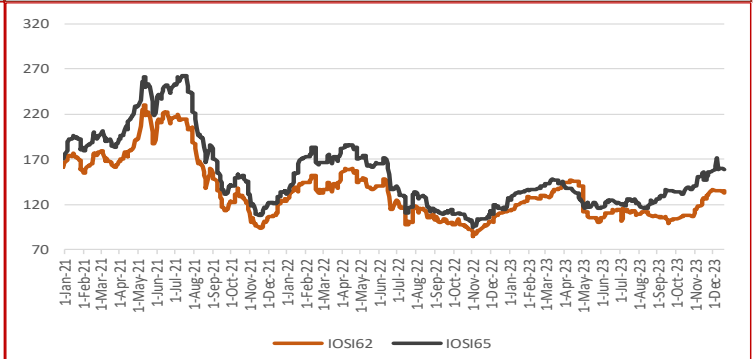
Week Ending Dec 15th, 2023	This week	Change %	Low [2]	High [2]
China Mines Concentrate Composite Index RMB/WT	1070.89	0.00%	706.36	1511.22

[1] Exchange rate applied: RMB/USD = 6.7522 [2] Last 12 months
[3] Weekly exchange rate applied: RMB/USD =6.87528

IRON ORE PORT INDEX, FOT QINGDAO (RMB/WT)



IRON ORE SEABORNE INDEX, CFR QINGDAO (USD/DMT)



IRON ORE PORT STOCK INDEX MONTHLY, QUARTERLY AND YEAR-TO-DATE AVERAGES

Dec 20th, 2023		FOT Qingdao (inc. 13% VAT), RMB/wet tonne							CFR Qingdao Equivalent (exc. 13% VAT), USD/dry tonne						
Index	Fe Content	August	September	October	November	MTD	QTD	YTD	August	September	October	November	MTD	QTD	YTD
IOPI62	62% Fe Fines	873	945	947	988	861	856	883	113.28	122.79	123.14	129.07	114.01	113.53	119.85
IOPI58	58% Fe Fines	759	838	869	884	765	760	767	99.13	109.67	113.94	116.42	101.99	101.52	104.49
IOPI65	65% Fe Fines	985	1057	1059	1100	978	974	1016	128.31	137.74	138.13	144.10	129.94	129.72	138.63

IRON ORE SEABORNE INDEX MONTHLY, QUARTERLY AND YEAR-TO-DATE AVERAGES

Dec 20th, 2023		CFR Qingdao, USD/dry tonne						
Index	Fe Content	August	September	October	November	MTD	QTD	YTD
IOSI62	62% Fe Fines	109.02	104.25	106.44	124.61	112.14	113.80	114.87
IOSI65	65% Fe Fines	119.61	132.40	135.69	150.46	122.86	126.97	128.26

FREIGHT RATES

Dec 19th, 2023		FREIGHT RATES - DRY BULK US$/wet tonne				
Route	Designation		Change	Change %	Low [2]	High [2]
W. Australia - Qingdao	C5	13.67	-0.63	-4.37%	3.57	16.77
Tubarao - Qingdao	C3	24.06	-0.33	-1.34%	6.70	36.40

IRON ORE PORT LUMP INDEX MONTHLY, QUARTERLY AND YEAR-TO-DATE AVERAGES

Week Ending Dec 15th, 2023		FOT Qingdao (inc. 13% VAT), RMB/wet tonne							CFR Qingdao Equivalent (exc. 13% VAT), USD/dry tonne [1]						
Index	Fe Content	August	September	October	November	MTD	QTD	YTD	August	September	October	November	MTD	QTD	YTD
IOPLI62	62.5% Fe Lump	951	1043	1113	1155	933	967	1017	118.96	130.25	139.38	145.07	119.96	121.92	134.98

IRON ORE INDEX PREMIUMS/DISCOUNTS

Dec 20th, 2023		PORT STOCK INDEX (RMB/WT)	
Index	Fe Content	Spread to IOPI62	% Spread to IOPI62
IOPI58	58% Fe Fines	-89	-8.88%
IOPI65	65% Fe Fines	112	11.10%

Dec 20th, 2023		SEABORNE INDEX (USD/DMT)	
Index	Fe Content	Spread to IOSI62	% Spread to IOSI62
IOSI65	65% Fe Fines	23.42	17.37%

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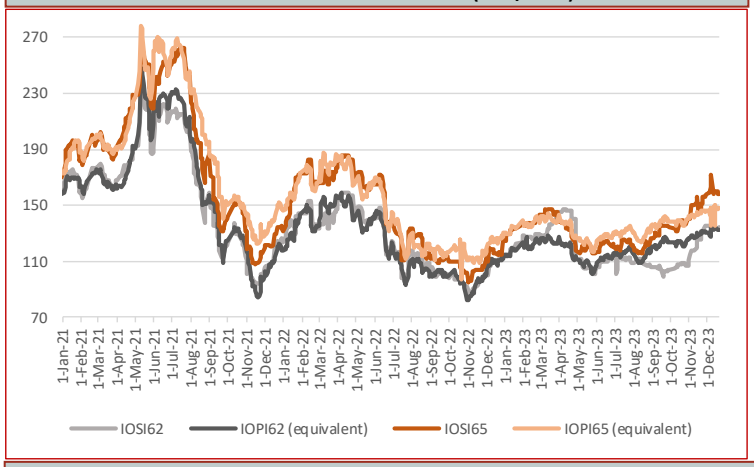
IRON ORE INDEX COMPARISONS (USD/DMT)



CHINA DOMESTIC COMPOSITE MINES INDEX (RMB/Wet Tonne, including VAT)

IRON ORE BRAND SPOT PRICE ASSESMENTS

Dec 20th, 2023	PORT STOCK INDEX (RMB/WT)		
	Price	Change	Diff to IOPI62
Roy Hill	949	4	-57
SIMEC Fines	881	4	-125
PB Fines	972	4	-34
Newman Fines	976	4	-30
MAC Fines	956	4	-50
Jimblebar Blended Fines	872	4	-134
Carajas Fines	1187	4	181
Brazilian SSF	973	4	-33
Brazilian Blend Fines	986	4	-20
RTX Fines	890	4	-116
West Pilbara Fines	917	4	-89

Dec 20th, 2023	SEABORNE INDEX (USD/DMT)		
	Price	Change	Diff to IOSI62
Roy Hill	130.30	2.60	-4.53
SIMEC Fines	126.80	2.60	-8.03
PB Fines	131.05	2.60	-3.78
Newman Fines	133.95	2.60	-0.88
MAC Fines	131.05	2.60	-3.78
Jimblebar Blended Fines	123.45	2.60	-11.38
Carajas Fines	164.35	2.60	29.52
Brazilian SSF	138.55	2.60	3.72
Brazilian Blend Fines	140.20	2.65	5.37
RTX Fines	124.65	2.58	-10.18
West Pilbara Fines	129.05	2.60	-5.78

Dec 20th, 2023	PORT STOCK INDEX (RMB/WT)		
	Price	Change	Diff to IOPI58
SSF	917	2	0
FMG Blended Fines	928	2	11
Robe River	929	2	12
Western Fines	933	2	16
Atlas Fines	925	2	8
Yandi	908	2	-9

IRON ORE INDEX NORMALISATION DIFFERENTIALS

Port Stock Index Product Differentials (RMB/wet tonne)

	Applicable range	Value	Change
1% Fe	High Grade Fe 60 - 63%	5.00	0.00
	High Grade Fe 63 - 64%	23.00	4.00
	High Grade Fe 64 - 65%	23.00	4.00
	High Grade Fe 65 - 65.5%	23.00	4.00
	Low Grade Fe	22.00	0.00
1% Alumina	High Fe Grade Al <2.25%	33.00	0.00
	High Fe Grade Al 2.25-4%	18.00	0.00
	Low Fe Grade Al <2.25%	78.00	3.00
	Low Fe Grade Al 2.25-4%	27.00	1.00
1% Silica	High Fe Grade Si <4%	19.00	2.00
	High Fe Grade Si 4-6.5%	20.00	-5.00
	Low Fe Grade	21.00	0.00
0.01% Phosphorus	High Fe Grade 0.09%<P<0.115%	9.00	1.00
	High Fe Grade 0.115%<P<0.15%	9.00	1.00
	Low Fe Grade 0.09<P<0.1%	5.00	0.00

Seaborne Index Product Differentials (USD/dry tonne)

	Applicable range	Value	Change
1% Fe	High Grade Fe 60 - 63%	1.75	-1.00
	High Grade Fe 63 - 64%	2.75	0.00
	High Grade Fe 64 - 65%	2.75	0.00
	High Grade Fe 65 - 65.5%	2.75	0.00
1% Alumina	High Fe Grade Al <2.25%	6.50	0.00
	High Fe Grade Al 2.25-4%	1.00	0.00
1% Silica	High Fe Grade Si <4%	3.25	0.50
	High Fe Grade Si 4 - 6.5%	3.25	0.50
0.01% Phosphorus	High Fe Grade 0.09%<P<0.115%	0.25	0.00
	High Fe Grade 0.115%<P<0.15%	1.00	-0.75

Port Stock Price Differentials to Qingdao Port for PB Fines (RMB/wet tonne)

Port	Value	Change	Port	Value	Change	Port	Value	Change	Port	Value	Change
Bayuquan	-100.00	0.00	Fangcheng	-25.00	0.00	Lanshan	0.00	0.00	Rizhao	0.00	0.00
Beilun	0.00	0.00	Jiangyin	-20.00	0.00	Lianyungang	0.00	0.00	Shekou	0.00	0.00
Caofeidian	-10.00	0.00	Jingtang	-10.00	0.00	Majishan	0.00	0.00	Taicang	-30.00	0.00
Dalian	0.00	0.00	Lanqiao	0.00	0.00	Qingdao	0.00	0.00	Tianjin	-40.00	0.00

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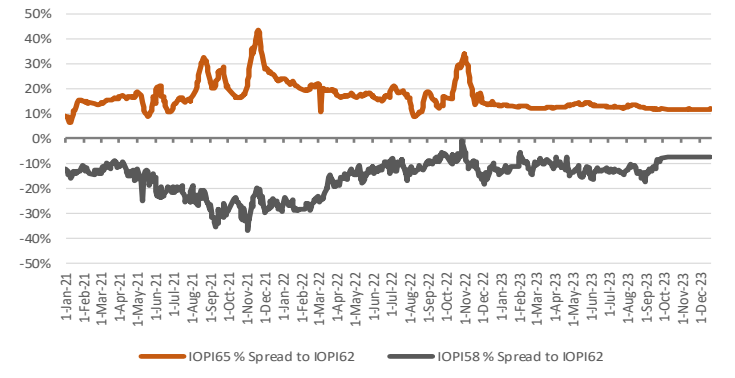
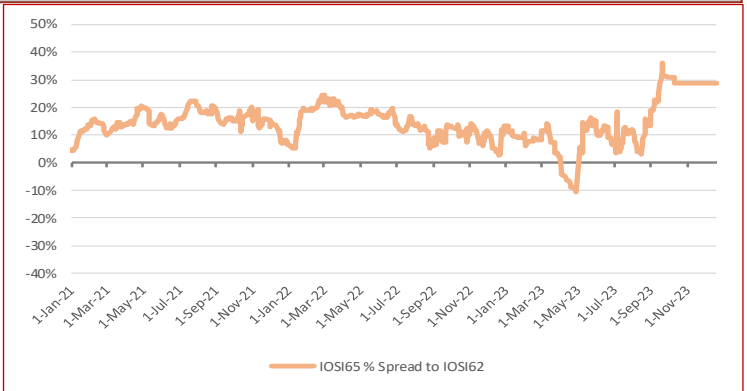
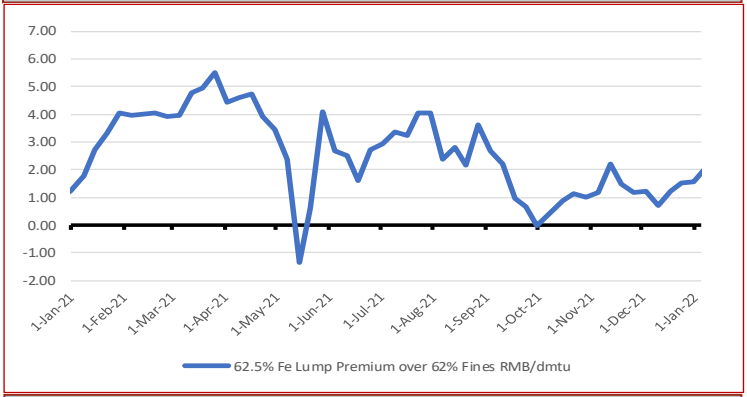
IRON ORE INDEX PREMIUMS/DISCOUNTS

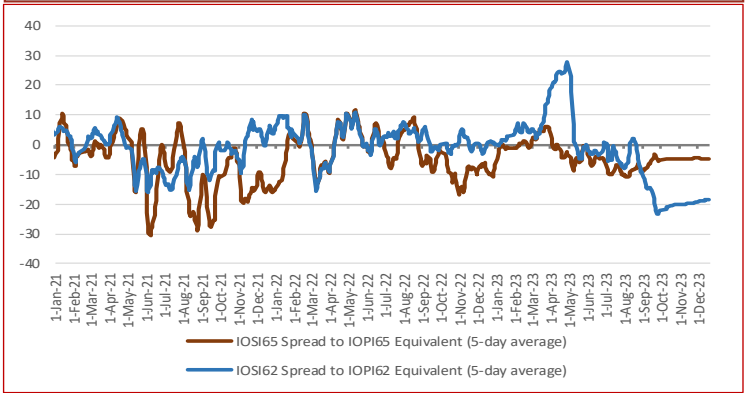




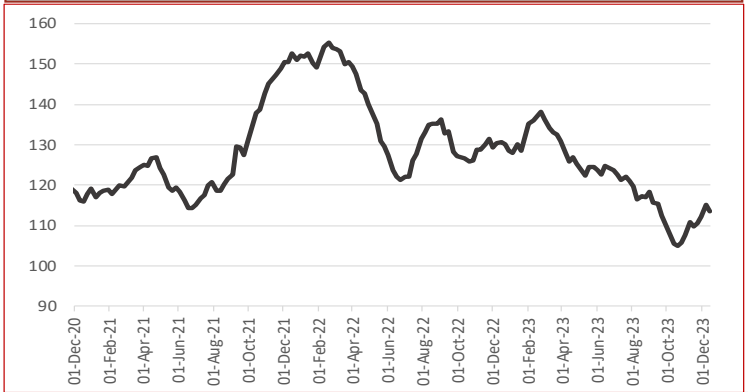
WEEKLY IRON ORE PORT STOCK LUMP PREMIUM (RMB/DMTU)



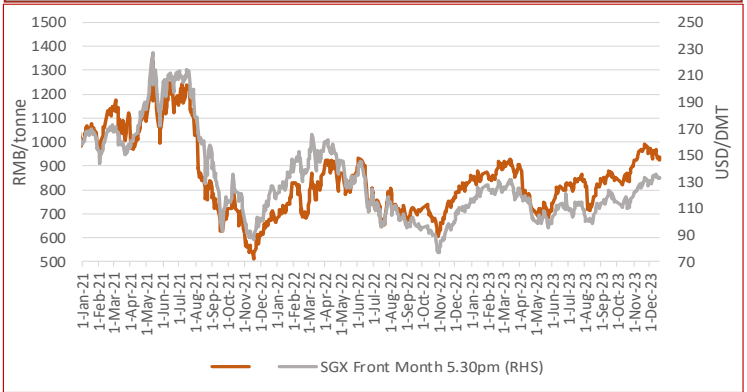
IRON ORE SEABORNE TO PORT STOCK PRICE SPREADS (USD/DMT)



TOTAL IRON ORE INVENTORIES AT CHINA PORTS (MILLION TONNES)



FUTURE TRADING—FRONT MONTH CLOSING PRICE



IRON ORE PORT INVENTORIES (MILLION TONNES)

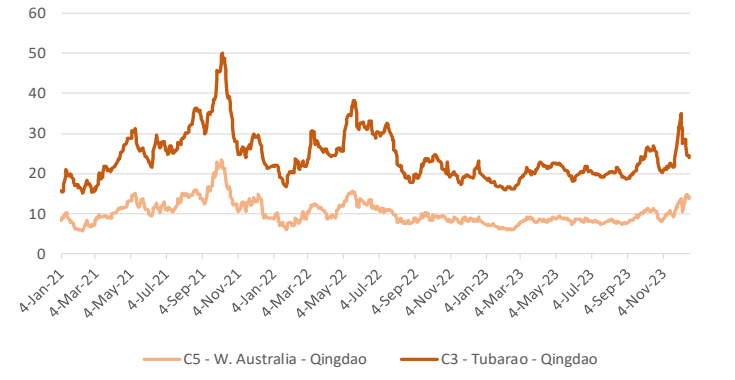
Week Ending Dec 15th, 2023

Province	This week	Change %	Low [2]	High [2]
Jingtang	8.77	-6.20%	8.41	19.20
Qingdao	24.09	0.75%	9.41	26.24
Caofeidian	10.76	4.06%	7.56	16.29
Tianjin	8.38	3.20%	6.49	12.97
Rizhao	13.48	1.20%	9.44	19.26
Total (35 Ports)	113.39	-1.49%	98.80	155.39

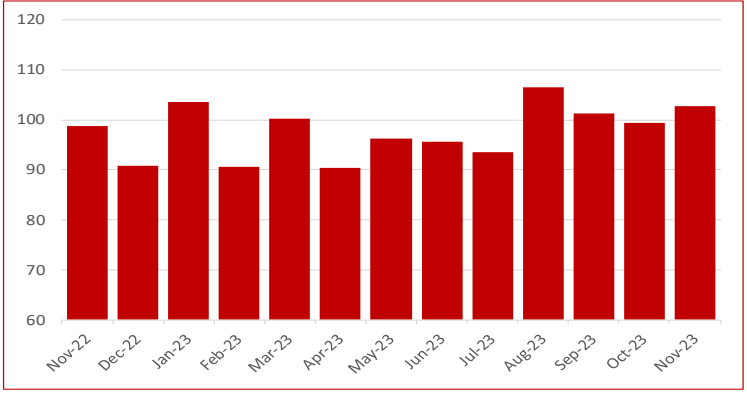
IRON ORE FUTURES CONTRACTS

	DCE (RMB/WMT)			SGX (USD/DMT)		
Closing Date	Dec 20th, 3pm close			Dec 20th, 5:30pm		
Contract	I2405	Change	Change %	Jan. 24	Change	Change %
Closing Price	939.00	16.00	1.73%	133.85	1.45	1.10%
Vol traded ('000 lots)	22.48	2.37	11.80%	6.12	-0.35	-5.41%
Open positions ('000 lots)	49.88	1.24	2.55%	32.06	-1.86	-5.48%
Day Low	923.5	3.50	0.38%	132.30	0.70	0.53%
Day High	945.0	12.00	1.29%	134.45	1.60	1.20%

DRY BULK FREIGHT RATES (USD/MT)



TOTAL CHINA IRON ORE IMPORT VOLUMES (MILLION TONNES)



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STEEL SPOT MARKET PRICES—CHINA

Steel Spot Market RMB/tonne

Product	2023/12/15	Change	Change %
ReBar HRB400 ϕ18mm	4,030	-60	-1.47%
Wirerod Q300 ϕ6.5mm	4,220	-70	-1.63%
HRC Q235/SS400 5.5mm*1500*C	4,040	-50	-1.22%
CRC SPCC/ST12 1.0mm*1250*2500	4,840	-20	-0.41%
Medium & Heavy Plate Q235B 20mm	4,030	-20	-0.49%
GI ST02Z 1.0mm*1000*C	5,040	-10	-0.20%
Colour Coated Plate	7,000	0	0.00%

CHINESE STEEL MILL PROFITABILITY

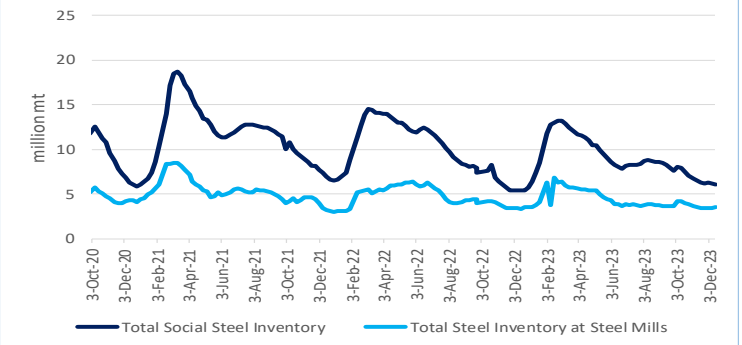
SMM Tracking of Steel Mill P&L - Rebar and Hot-rolled Coil (RMB/tonne)

Category	Price	Change (WoW)	Note
MMi (Fe 62%), USD/mt exluding tax	134.69	-1	Mmi CFR Equivalent index for 1st Feb
Coke	2,780	110	2nd grade met coke, Tangshan, incl. tax
Steel Scrap	2,910	0	steel scrap (6mm) in Zhangjiagang, exl. tax
Billet Cost	3,440	58	Q234, incl. tax
Rebar cost - Blast furnace	4,114	65	calculated based on theoretical weight, incl. tax
Rebar profit - Blast furnace	-184	-165	based on Shanghai prices, incl. tax
Hot-rolled coil cost - Blast furnace	4,179	66	based on actual weight, incl. tax
Hot-rolled coil proft - Blast furnace	-169	-136	based on Shanghai prices, incl. tax

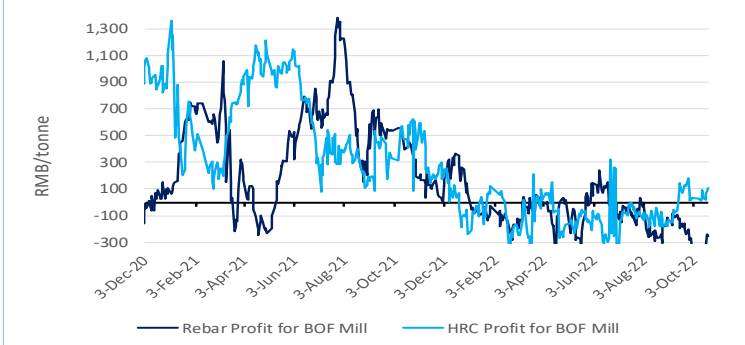
Note: 1. Costs in the table are caluclated based on todays market prices and facout our management, sales, financial and depreciations fe

2. The cost refers to average cost in the industry based on SMM's survey of small, medium and large mills in China

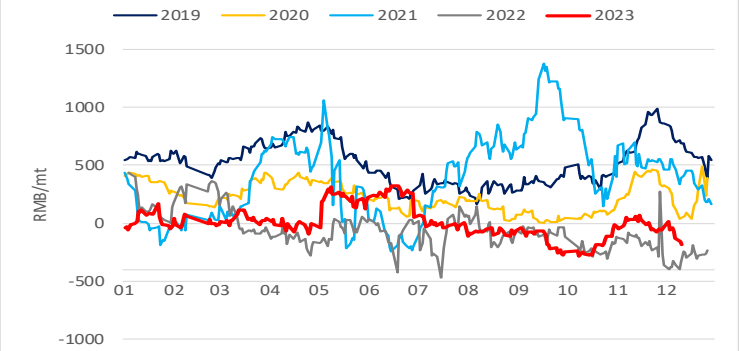
CHINESE STEEL INVENTORIES



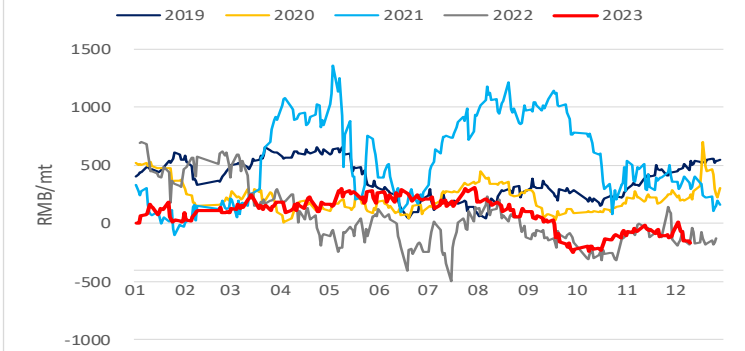
CHINESE STEEL MILL PROFITABILITY



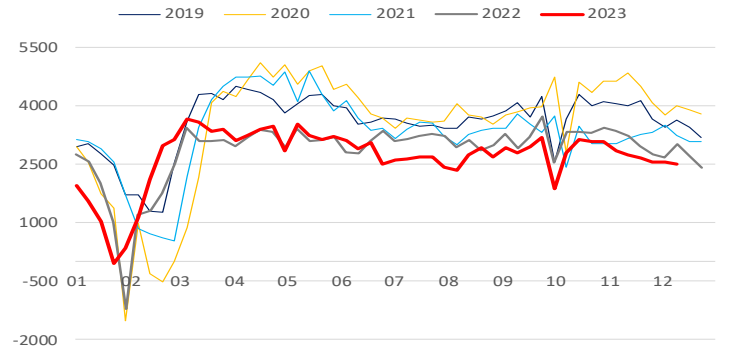
CHINESE STEEL MILL PROFITABILITY—Rebar



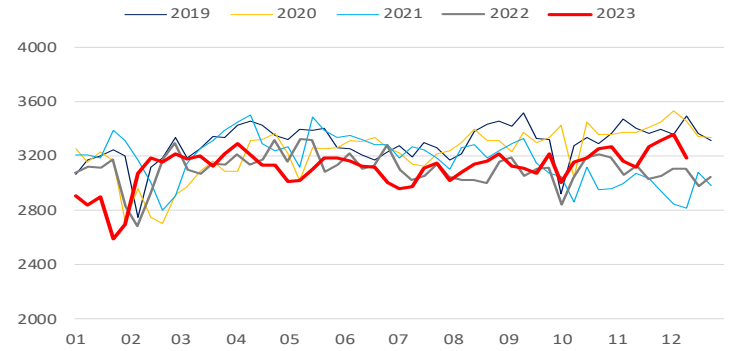
CHINESE STEEL MILL PROFITABILITY—Hot-rolled Coil



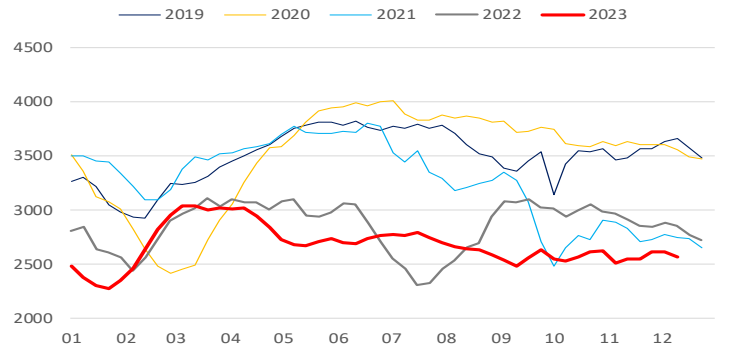
CHINESE STEEL CONSUMPTION—Rebar



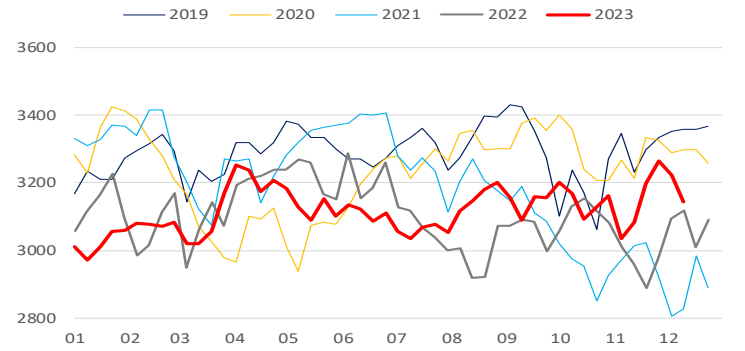
CHINESE STEEL CONSUMPTION—Hot-rolled Coil



CHINESE STEEL PRODUCTION—Rebar



CHINESE STEEL PRODUCTION—Hot-rolled Coil



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IRON ORE INDEX SPECIFICATIONS, COMPILATION RATIONALE AND DATA EXCLUSIONS

Iron Ore Index Specifications (Port and Seaborne)

	65% Fe Fines	62% Fe Fines	58% Fe Fines	62.5% Fe Lump
Fe %	65.00	62.00	58.00	62.50
Alumina %	1.40	2.25	2.25	1.50
Silica%	1.50	4.00	5.50	3.50
Phosphorus %	0.06	0.09	0.05	0.08
Sulphur %	0.01	0.02	0.02	0.02
Moisture %	8.00	8.00	9.00	4.00
Sizing	Granular size below 10mm for at least 90% of cargo; maximum of 40% below 150 micron			Size below 6.3mm max 15% Size above 31.15mm max 25%
Pricing Point	Qingdao Port (FOT and CFR respectively)			FOT Qingdao Port
Timing (Seaborne)	Loading within 4 weeks, Delivery within 8 weeks			Delivery within 2 weeks
Payment Terms	L/C at sight			L/C at sight or CAD

Iron Ore Index Compilation Rationale and Data Exclusions

MMi iron ore indices are compiled from data provided by companies that are part of the iron ore supply chain and involved in spot market transactions. The indices are calculated using detailed transaction-level data submitted to MMi by these companies. This data is normalised to the appropriate specifications and screened to remove outliers before volume-weighted average prices are calculated from the remaining core set of data.

For more details on MMi's iron ore methodology please download the guide published on our website at: www.mmiprices.com

Data Exclusions*

	62%	58%	65%
Port Index	0	0	0
Seaborne index	0		0
Lump Index 62.5	0		

* Number of price submissions for iron ore indices that were excluded from index calculations today as they were anomalous and could not be verified

IRON ORE DOMESTIC CONCENTRATE INDEX CALCULATION METHODOLOGY

The compilation method for price index generally refers to the compilation method of CPI price index and other price indices, breakdown the price data and calculate the average value according to a certain method, taking the vertical axis as the regional composite index (average of different grade index) and the horizontal axis as the grade composite index (average of different regional index), a total composite index for domestic ore can be output ultimately. The process system is also adopted in the calculation i.e. each sub-index can be obtained as well.

AVERAGE IRON ORE SPECIFICATIONS APPLIED FOR BRAND PRICE ASSESSMENTS

PORT STOCK BRANDS

Dec 20th, 2023	Specifications applied for 62% brand assessments				
	Fe	Alumina	Silica	Phos	Moisture
Roy Hill	60.88%	2.35%	4.45%	0.055%	9.26%
SIMEC Fines	60.00%	2.30%	6.30%	0.060%	6.00%
PB Fines	61.64%	2.31%	3.87%	0.099%	9.45%
Newman Fines	62.49%	2.31%	3.98%	0.090%	7.72%
MAC Fines	60.43%	2.38%	4.64%	0.078%	7.57%
Jimblebar Blended Fines	60.87%	2.95%	4.53%	0.111%	7.32%
Carajas Fines	64.90%	1.39%	1.82%	0.080%	7.78%
Brazilian SSF	62.00%	1.00%	6.50%	0.040%	6.00%
Brazilian Blend Fines	62.56%	1.58%	4.79%	0.091%	8.72%
RTX Fines	61.00%	3.10%	4.50%	0.135%	7.50%
West Pilbara Fines	60.10%	2.30%	4.70%	0.075%	8.50%

Dec 20th, 2023	Specifications applied for 58% brand assessments				
	Fe	Alumina	Silica	Phos	Moisture
SSF	56.49%	3.20%	6.19%	0.065%	9.18%
FMG Blended Fines	58.20%	2.43%	5.54%	0.057%	8.29%
Robe River	56.44%	3.16%	5.73%	0.042%	8.44%
Western Fines	57.88%	2.87%	7.50%	0.062%	7.45%
Atlas Fines	58.00%	1.85%	5.50%	0.090%	9.00%
Yandi	56.87%	1.58%	6.41%	0.042%	9.53%

SEABORNE BRANDS

	Specifications applied for 62% brand assessments				
	Fe	Alumina	Silica	Phos	Moisture
Roy Hill	60.70%	2.30%	4.90%	0.055%	8.00%
SIMEC Fines	60.00%	2.30%	6.30%	0.060%	6.00%
PB Fines 62%	62.00%	2.60%	4.30%	0.090%	9.50%
Newman Fines	62.30%	2.40%	4.30%	0.095%	8.00%
MAC Fines	60.80%	2.35%	4.40%	0.075%	7.80%
Jimblebar Blended Fines	60.50%	3.10%	4.80%	0.120%	7.80%
Carajas Fines	65.10%	1.50%	1.70%	0.080%	8.50%
Brazilian SSF	62.00%	1.00%	6.50%	0.040%	6.00%
Brazilian Blend Fines	63.00%	1.50%	5.00%	0.070%	7.00%
RTX Fines	61.00%	3.10%	4.50%	0.135%	7.50%
West Pilbara Fines	60.10%	2.30%	4.70%	0.075%	8.50%

BLOOMBERG TICKERS

PORT STOCK INDICES

	FOT Qingdao (RMB/wet tonne)	CFR Qingdao Equivalent (USD/dry tonne)
IOPI62	IRCNQ001	IRCNQ004
IOPI58	IRCNQ002	IRCNQ005
IOPI65	IRCNQ003	IRCNQ006
IOPLI62	IRCN0036	IRCN0037

SEABORNE INDICES

	CFR Qingdao (USD/DMT)
IOSI62	IRCN0034
IOSI65	IRCN0035

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